

Defect Management 16_pms_defect_mgmt

Defect Management

1 Formal Description

Identifier	Name	Owner	User ID
16_PMS_DM	DM Process	Hiromi.Kato@alpsalpine.com	B0952028

Process Designers

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1.1 Scope for Application of this process

This process is focusing Defect Management only. This process is covered for following phase.

- Instration phase
- Developing phase (WK / PR / PP / AP)
- Production and maintenance phase (AP / MP)
- Field service phase(MP)

NOTE:

Other “problems (Issue)” are managed in other “MAN3” processes:

- Project management: LOP
- Quality Assurance: Audit Finding etc.
- Any problem which is obstructing the development process

1.2 Tailoring Guideline

Defect management process to be tailored based on the project phase. Also, if there is some project specific item, it should be considered as a tailoring item.

* **Project phase** (SW Domain Project)

- Architecture Phase (Project preparation phase)
- Implementation Phase (AKA: Construction Phase)
 - Maturity Phase (AKA: Bug-fixing Phase)
 - Stability-Phase (AKA: Show-stopper Phase, which is too negative)
- Production and maintenance phase
- Field Service phase

* **Project specific**

i.g.

- Synchronization condition with external systems. (with Customer and/or SubSupplier)
- Other system information (Read-only)

1.3 Relations

- Relations
 - Policies
 - Standards
 - A-SPICE 3.1
 - Covering: [SUP.9 Problem Resolution Management](#)
 - Partly covering: [MAN.3 Project Management](#)
 - Partly covering: [SWE.4 Software Unit Verification](#)
 - Partly covering: [SWE.5 Software Integration and Integration T](#)
 - Partly covering: [SWE.6 Software Qualification Test](#)
 - Partly covering: [SYS.4 System Integration and Integration Test](#)
 - Partly covering: [SYS.5 System Qualification Test](#)
 - Partly covering: [ACQ.4 Supplier Monitoring](#)
 - IATF 16949
 - e.g. Chapter 2-3
 - ISO 26262
 - If functional Safety requirement(ASIL) are included in project scope, execute process which is defined in [FSQSX\[English\]](#) [FSQSX\[Japanese\]](#)
 - ISO/SAE21434
 - If Cyber Security requirement is included in project scope, please see [12 Cyber Security](#).
 - Unified QS
 - [SUP.9 Problem Resolution Management](#)
 - [MAN.3 Project Management](#)
 - [ACQ.4 Supplier Monitoring](#)

2 Strategy

2.1 Purpose

- Defect Management is for the following purposes:
 - Ensure that problems are identified, analyzed, managed, and controlled until they are resolved.
 - Record problems, analyze trends, and identify root causes
- Defect Managementは、下記を目的とする
 - 問題が識別され、分析され、管理され、解決まで制御されることを確実にすること。
 - 問題記録し、傾向分析、根本原因の特定を行うこと。

2.2 Objectives

- Introduce Defect Management System. All
- 問題管理システムの導入。すべての問題は

the defect is controlled by the management system.

- The communication for the defect issue is controlled in the defect management system as much as possible.
- All the defect is managed by each Domain. To analyze all the feature and the problem for the each domain to effective defect management.
- To manage the all defect ticket based on the development process until the modification is completed.
- Transparent the analysis result, rootcause, countermeasure and the test result to all stakeholder.

管理システムによって制御する。

- 不具合問題の通信は、問題管理システムにおいて可能な限り制御する。
- すべての問題は各ドメインによって管理する。各ドメインに関するすべての機能と問題を問題管理にて分析する。
- 修正が完了するまで開発プロセスに基づいてすべての問題チケットを管理する。
- 分析結果、根本原因、対策、テスト結果を全ての利害関係者に対して透過的に行う。

2.3 Strategy

- Establish defect guideline.
 - continuous improvement for defect guideline.
- Develop a Defect management strategy.
- Trace-ability with other ticket.(LOP, HLF, CR, Task...etc)
- Decide infrastructure (Tool).
 - Design synchronization rules with external defect management system.
 - Trace storage and synchronization management.
- Identify and record the defect.
- Track the defect to closure.
- Dashboard Management.
 - Defect curve management.
 - Trend analysis.
 - Inflow/Outflow count.
 - TAT.
 - Burndown chart.

3 Stakeholders

3.1 Skills, Capabilities, Responsibility & Authority

 **To describe here:** which main roles for this process do exist and what is their general responsibility with regard to the overall process. For each role the generic Skills and Capabilities shall be defined. The definition of the roles is part of the EA/Activity Model

Role	Skills	Capabilities	Responsibility	Authority
Defect manager	1) Common language with Customer and Supplier	1) Coordinate with project member 2) Coordinate with the Infrastructure for Defect management tools.	1) Understand the Defect Management requirement 2) Confirm the unclear point in Defect management 3) Establish the defect management guideline 4) Report the Defect management status to stakeholders	1) Create the defect management plan. 2) Create the defect management guideline 3) Create the dashboard contents 4) Defect ticket operation support.
Project infrastructure	1) System Infrastructure knowledge 2) Common language with Customer and Supplier	1) Expert knowledge of System infrastructure development	1) Develop the System infrastructure for defect management	1) Set up the defect management tools (Synchronization with Customer / supplier)
Validation Domain	1) Common language with Customer and Supplier	1) Knowledge of test environment, Test case creation	1) Review the test case and the test result 2) Review the test environment 3) Report the test result (Review the test status)	1) Create the test case 2) Create the stability KPI calculation and report 3) Validate the defect ticket.
Project Domain Leader (SW, HW, Mech)	1) Common language with Customer and Supplier	1) Make a communication with customer FO	1) Review and approve the rootcause / countermeasure from Engineer.	1) Create the analysis report
Engineer (SW, HW, Mech)	1) Common language with Customer and Supplier	1) Knowledge of analysis	1) Analysis the defect behaviour 2) Trace analysis 3) Find the root cause 4) Design the modification	1) Update the defect status to the defect ticket
Production Engineering Department	1) Common language with Customer and Supplier	1) Knowledge of analysis 2) Knowledge of the test case in the production line	1) Analysis the defect behaviour 2) Coordinate the defect analysis to Domain project leader	1) Create the test case at the production line
Quality Engineering Department	1) Common language with Customer and Supplier	1) Coordinate with Project Manager and Customer	1) Coordinate the defect information from Customer 2) Coordinate the defect analysis to Domain project leader	2) Report the analysis status to stakeholder

X = Responsibility (x) = possibility

AlpsAlpine production event which the defect was found		WK / PR / PP / AP			AP / MP	MP	
Responsibility to investigate / fix the defect		In developing process	In production process	QC / DVT / PVT	0km (APN factory)	0km (Customer factory)	Field claim
Software Engineer Department	Unit Software	X	X	X	X	X	X
HW Engineer Department	Unit Hardware	X	X	X	X	X	X
Mech Engineer Department	Unit Hardware	X	X	X	X	X	X
Validation Engineering Department	- Test program - Test Environment	X					
Production Engineering Department	Line and Jig : - Inspection program - Inspection jig - Inspection method - Production process	X	X	X	X		
Quality Engineering Department	Inspection program Inspection jig	X (x)		X			

3.2 Stakeholder involvement

Meeting	Method /Type	Frequency	Defect Manager	Project Domain Leader (SW, Elec, Mech)	Validation Domain	Project Manager
Defect workflow / guideline	WS with Stakeholders	Several times	AR	S	S	A
Defect ticket review	F2F MTG, Web Conference	Weekly	AR	AR	S	A
Project MTG	F2F MTG, Web Conference	Weekly	I	R	S	AR
Lessons Learn	F2F MTG, Web Conference	SW release timing	R	R	R	A
	<u>Responsible, Approving Authority, Support, Informed and Consulted</u>					

4 Input Work products

- Project SoW
- Defect Tickets handling management

- プロジェクトSoW
- 問題チケットの取り扱い管理

- (including from customer and sub supplier)
- Log-, Trace data storage management
- Synchronization status check (with Customer and Sub Supplier)
- ログ、トレースデータのストレージ管理
- 問題ステータス確認

5 Output Work products

- Defect Management Plan
- Defect Management Plan Review Record
- Defect Management Status Report
- Defect Guideline (typically wiki)
- Defect Dashboard
- Ticket operation(handling) support
- 問題解決管理計画書
- 問題解決管理計画書レビュー記録
- 問題管理ステータス報告書
- 問題ガイドライン
- 問題ダッシュボード
- チケット操作(取り扱い)サポート

6 Process Practices

6.1 Life Cycle relation

6.2 Sequence & interaction with other processes

A tight relationship exists with the Project Management process.

6.3 Entry & Exit conditions (triggers)

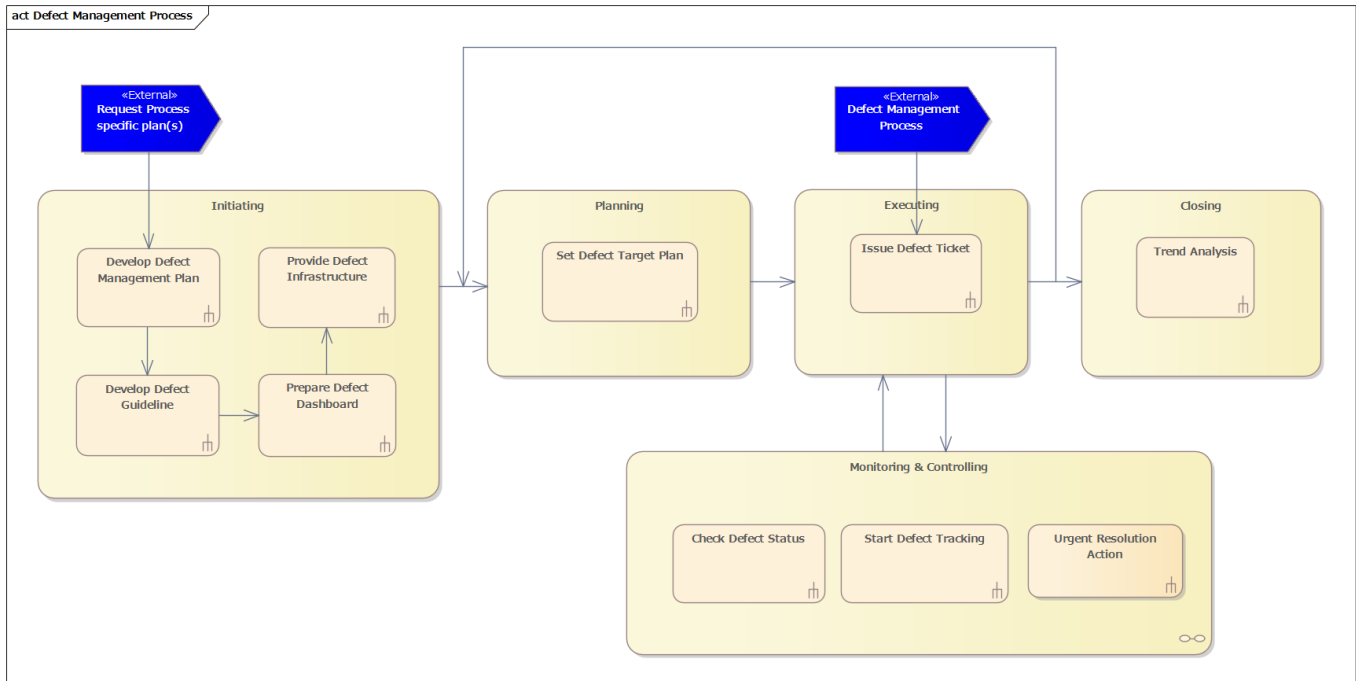
- **Start Conditions**
 - When the requirement functions are implemented.
 - When the HLFs are matured.
 - When the key features are implemented.
 - When the CRs are implemented.
 - When the Defect ticket is fixed.
 - ..
- **Exit Conditions**
 - SOP development is completed.
 - Maintenance phase is completed.
 - Service phase is completed.

* **Note** This “Exit Conditions” are based on the contract with Customer.

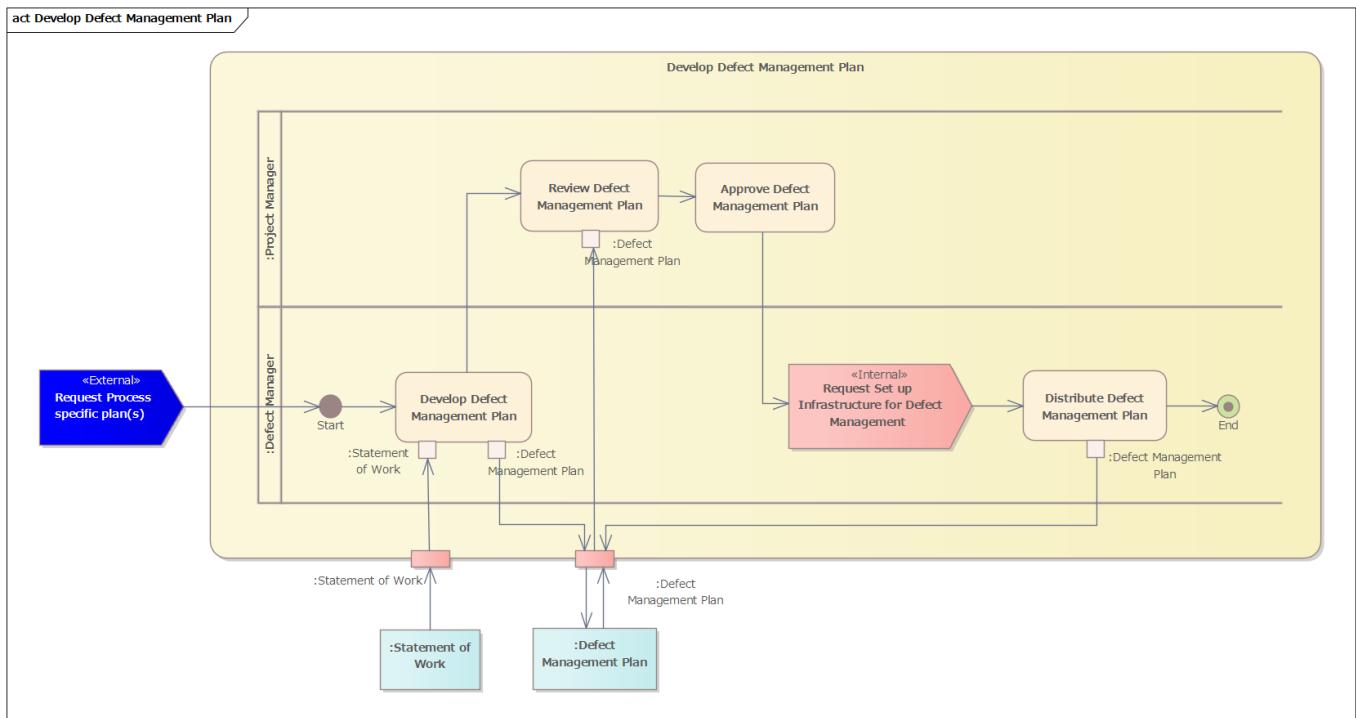
6.4 Process Flow

6.4.1 Defect Management Process Overview

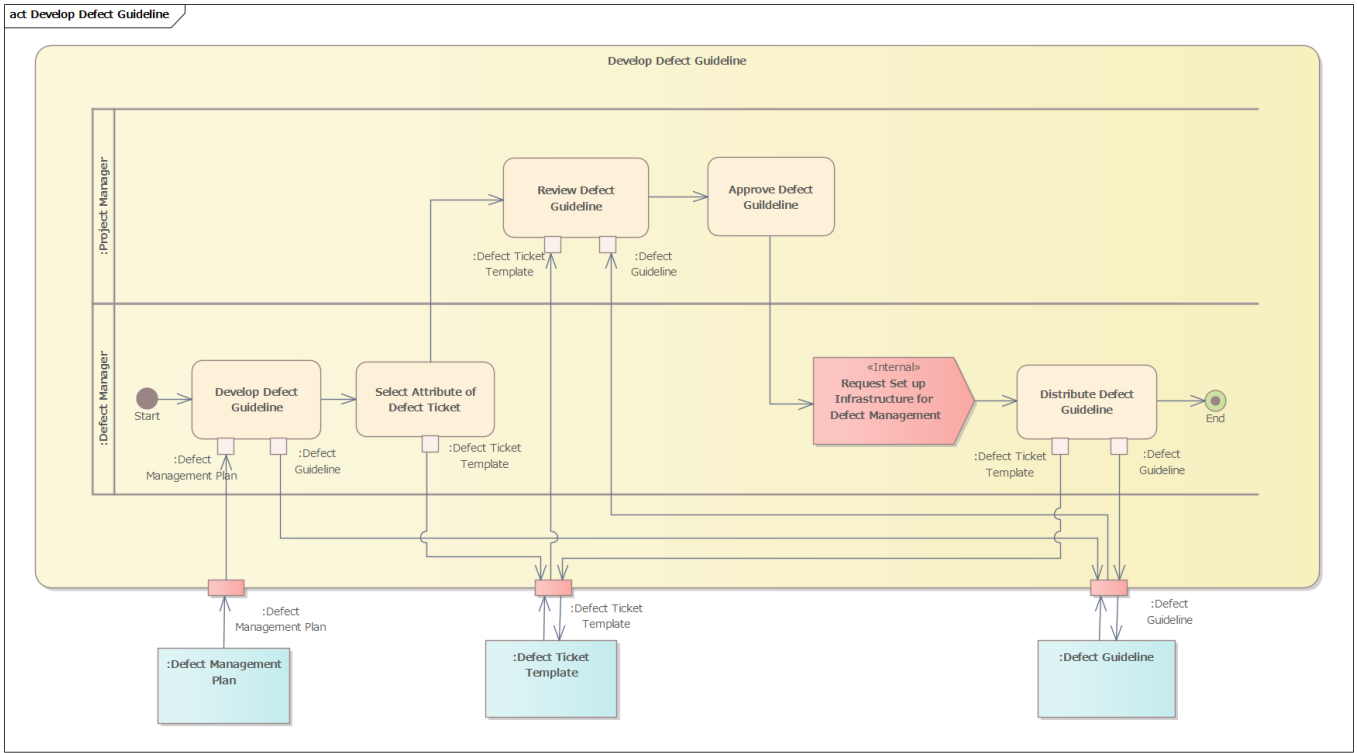
Please refer the **Defect ticket flow** for actual defect management process.



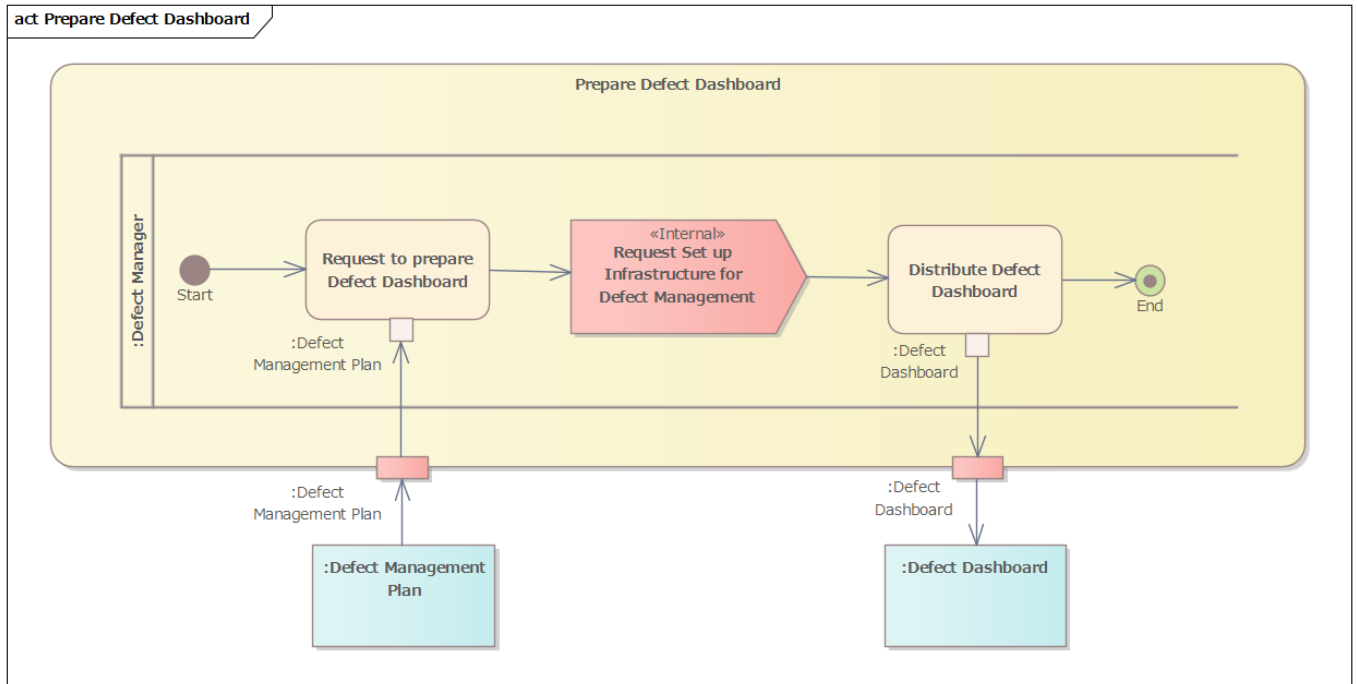
6.4.2 Develop Defect Management Plan



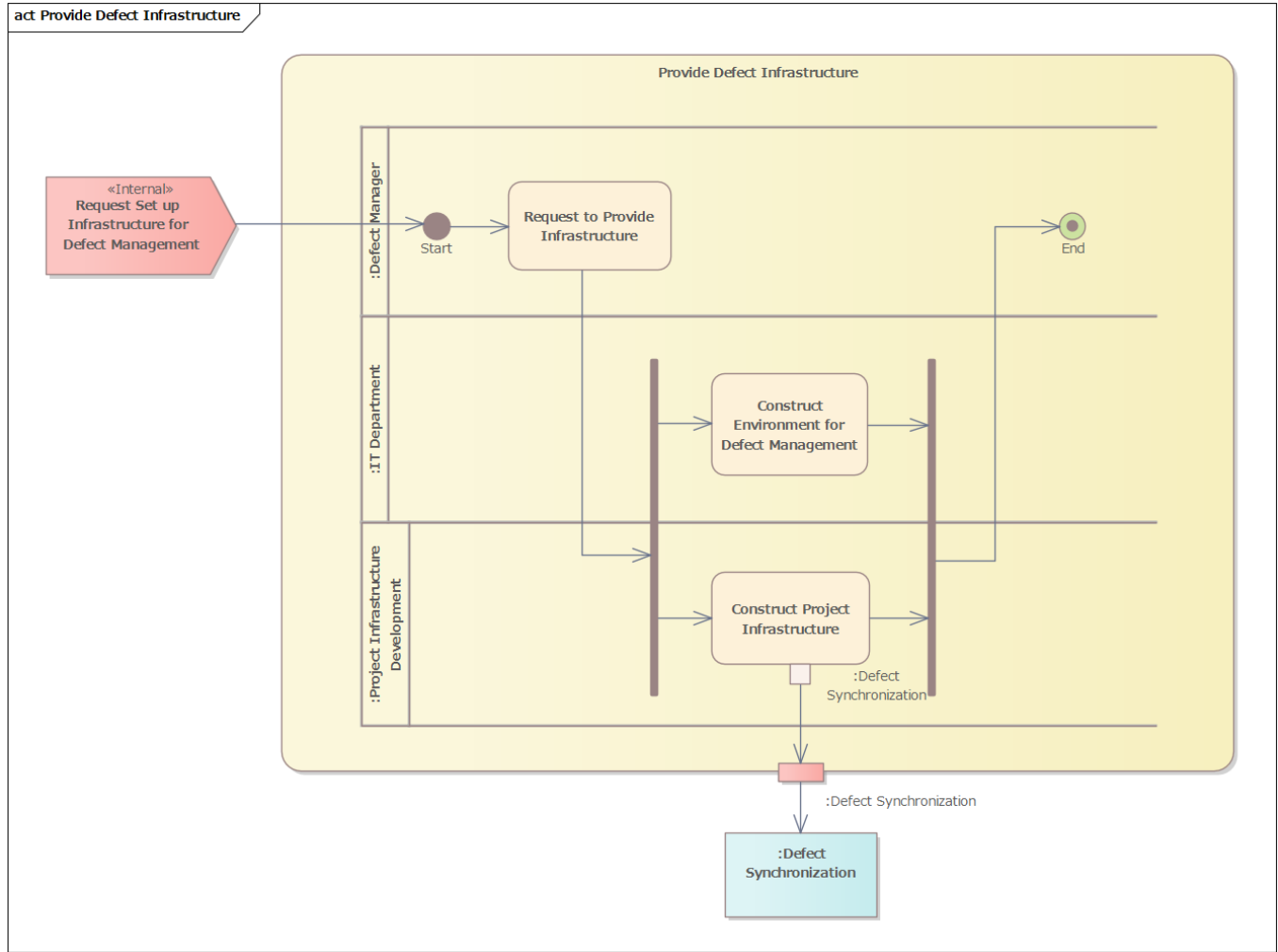
6.4.3 Develop Defect Guideline



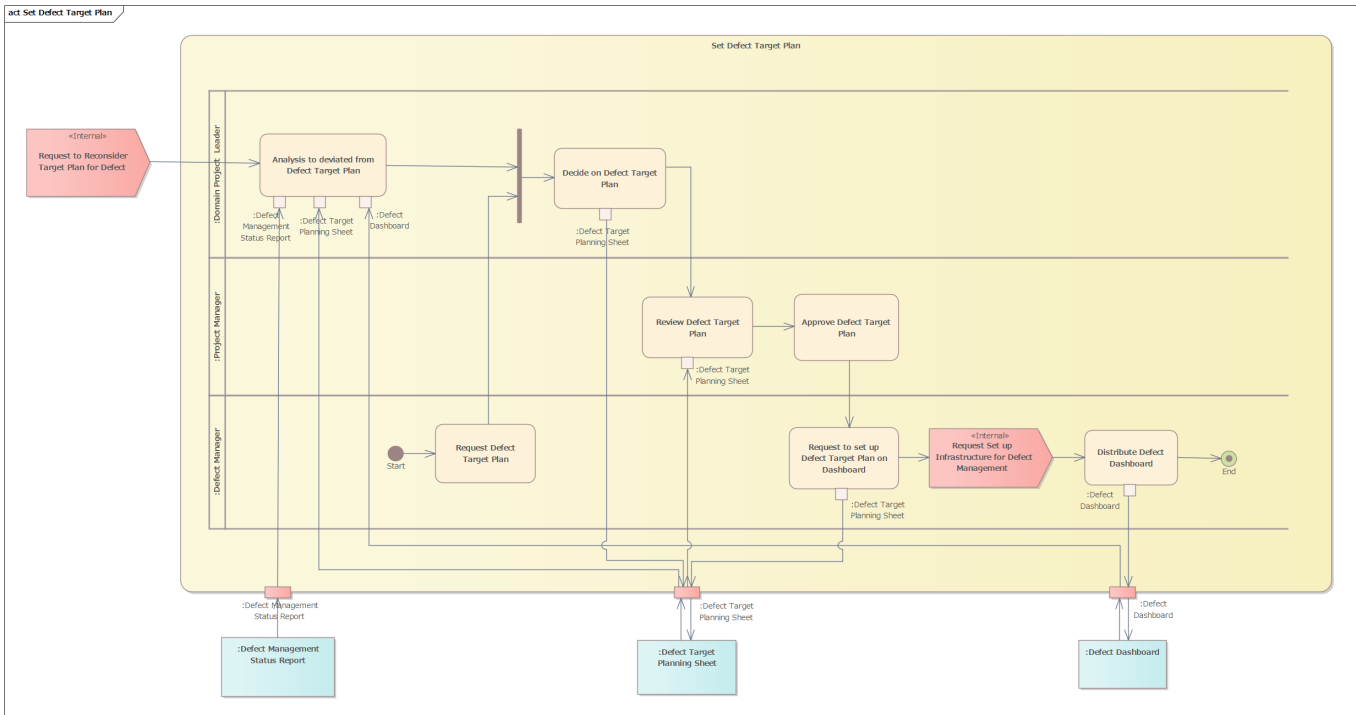
6.4.4 Prepare Defect Dashboard



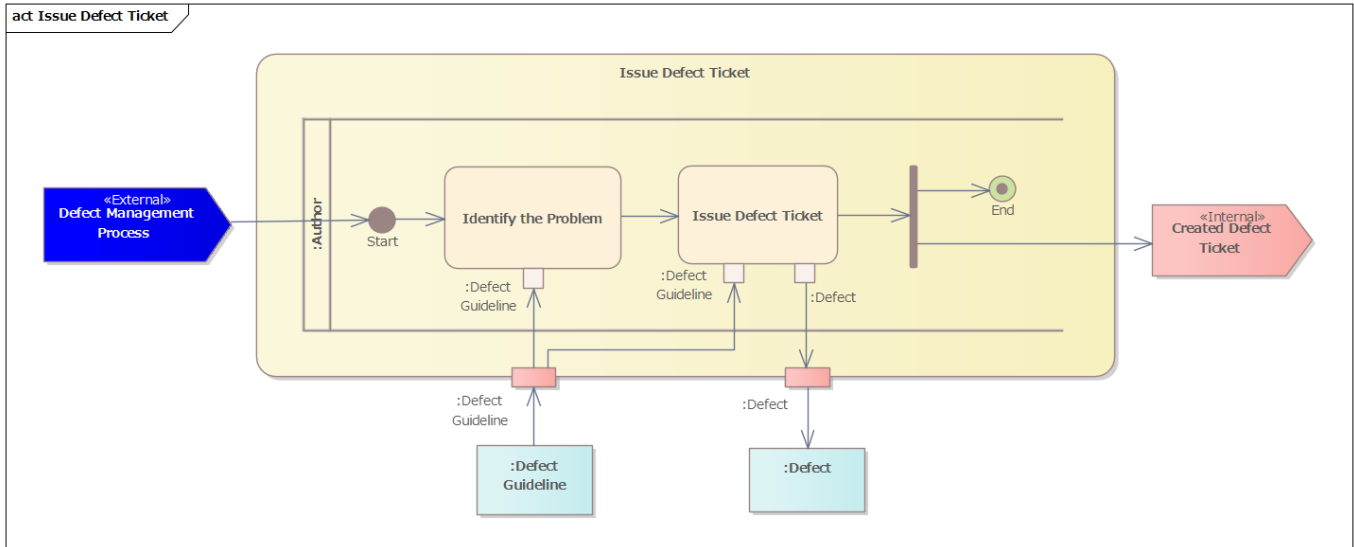
6.4.5 Provide Defect Infrastructure



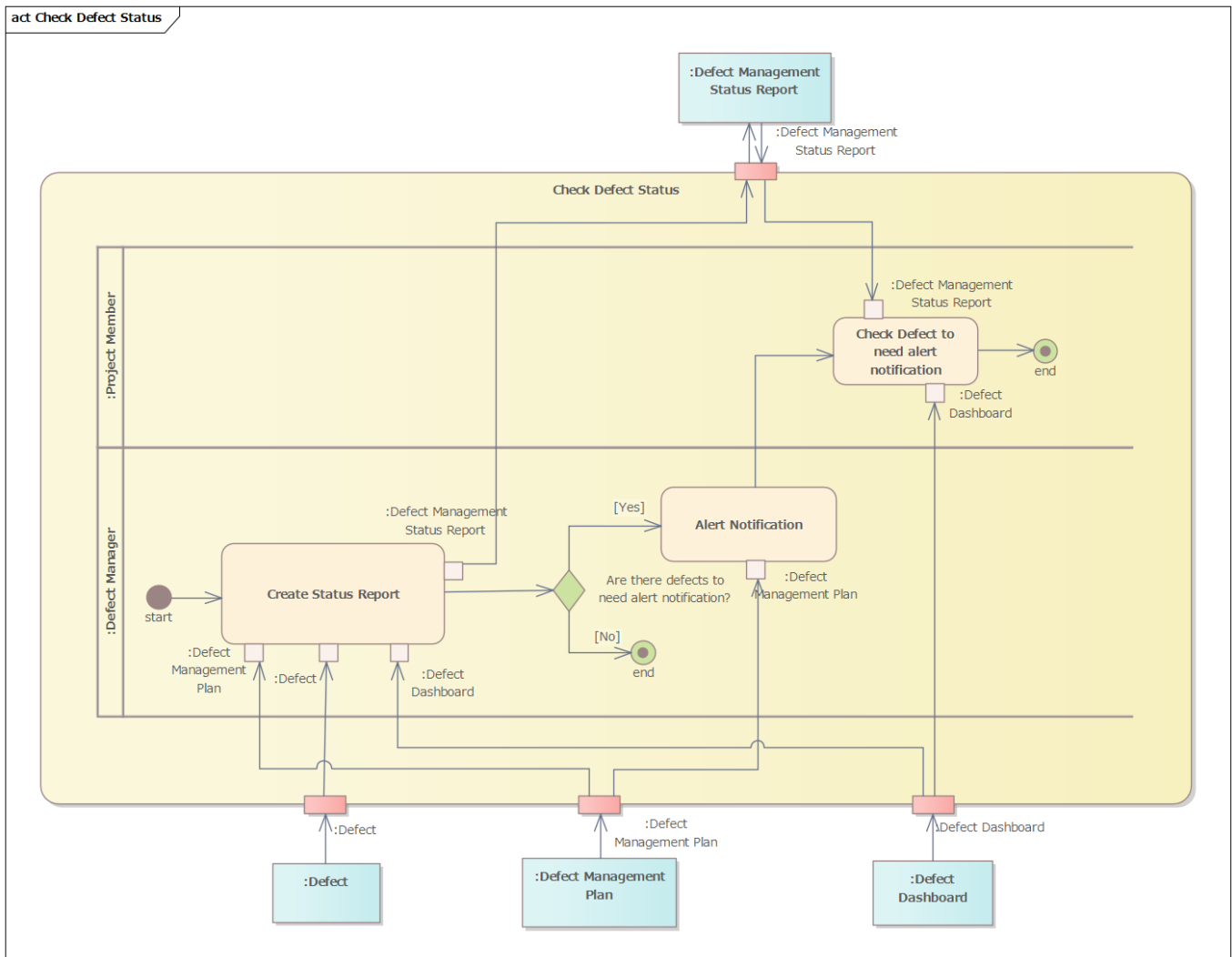
6.4.6 Set Defect Target Plan



6.4.7 Issue Defect Ticket

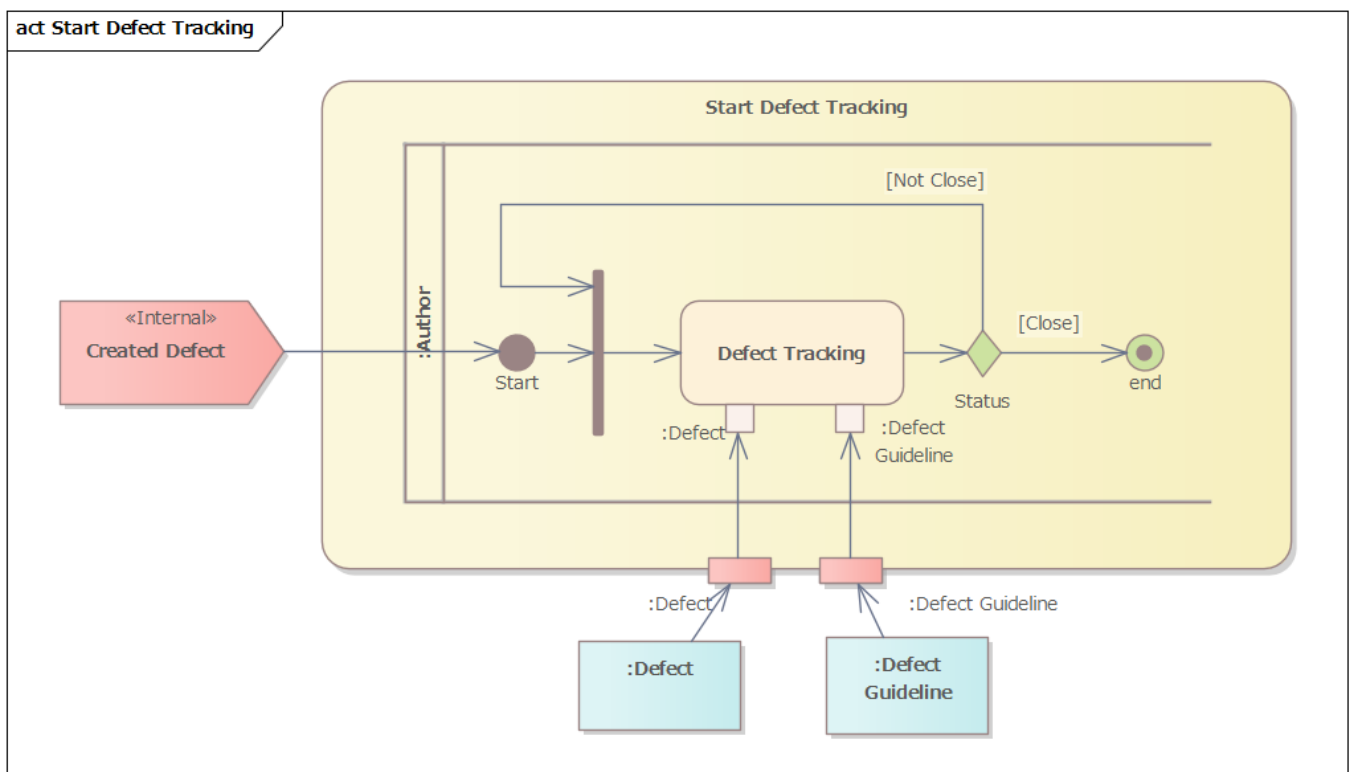


6.4.8 Check Defect Status

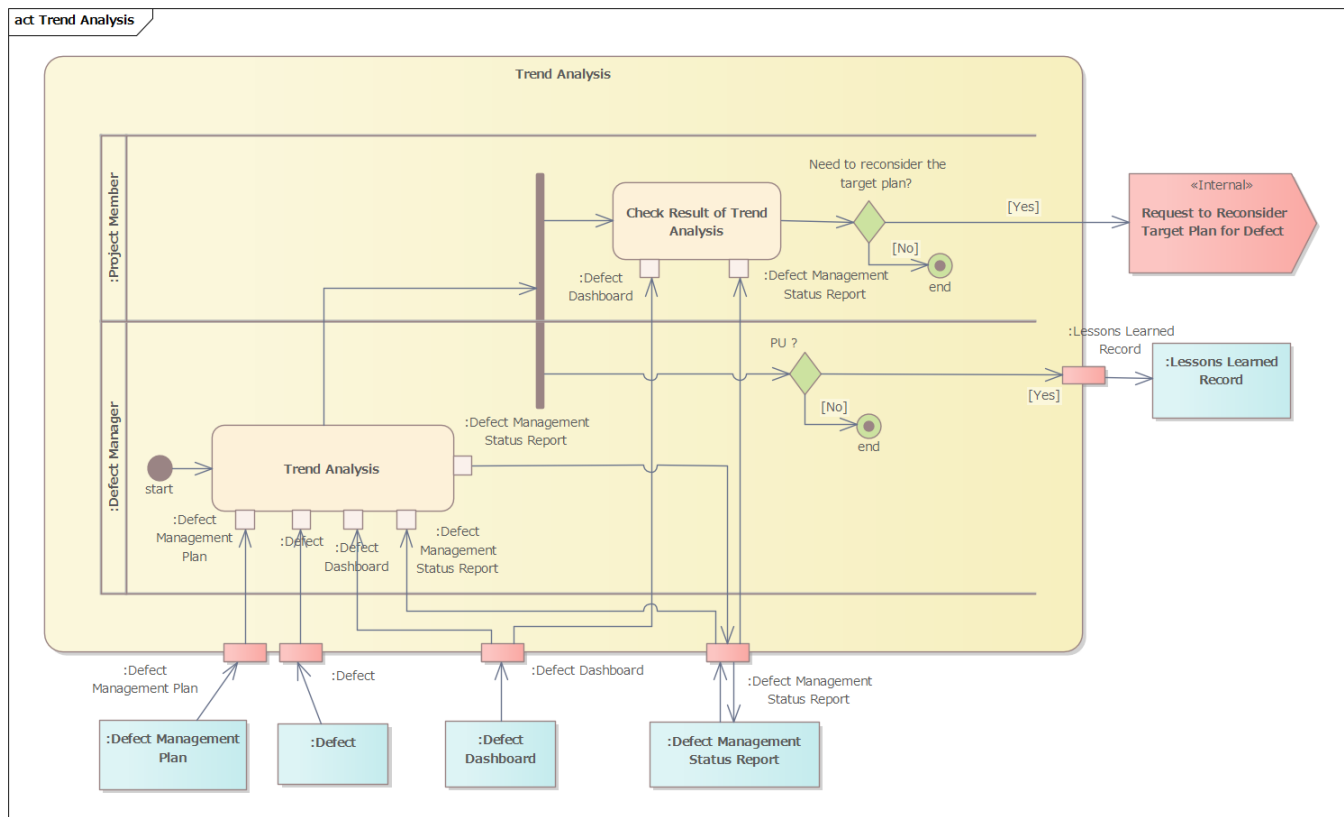


- Alert notification
 - Defect Manager notifies project members of the alert, when defects that is alerted target occurs.
 - ex.) The alerted target
 - Affect other system or stakeholders
 - Date of Defect ticket is overdue
 - TAT of Defect ticket is longer
 - The alert is notified by one of the following measures.
 - report at a regular meeting
 - send by E-mail
 - Alert notification target person
 - Report destination varies depending on the type of problem. The mailing list is prepared in advance.
 - All Project stakeholders with customers
 - Developer , validation members only
- アラート通知
 - アラート対象となる問題が発生した場合、アラート通知を行う。
 - 例) アラート対象となる問題
 - 他のシステムまたは、関係者に影響を及ぼす可能性がある問題
 - Defect チケットの期限がきれている
 - TATが長いDefect チケット
 - アラートを通知する方法は、以下のいずれかの手段を使う。
 - 定例MTGでの報告
 - メールでの報告
 - アラート通知対象者
 - 問題の種類によって報告先は変わります。予めメーリングリスト等の用意をします。
 - 顧客を含むすべてのプロジェクト関係者
 - 開発者、評価担当者のみ

6.4.9 Start Defect Tracking



6.4.10 Trend Analysis



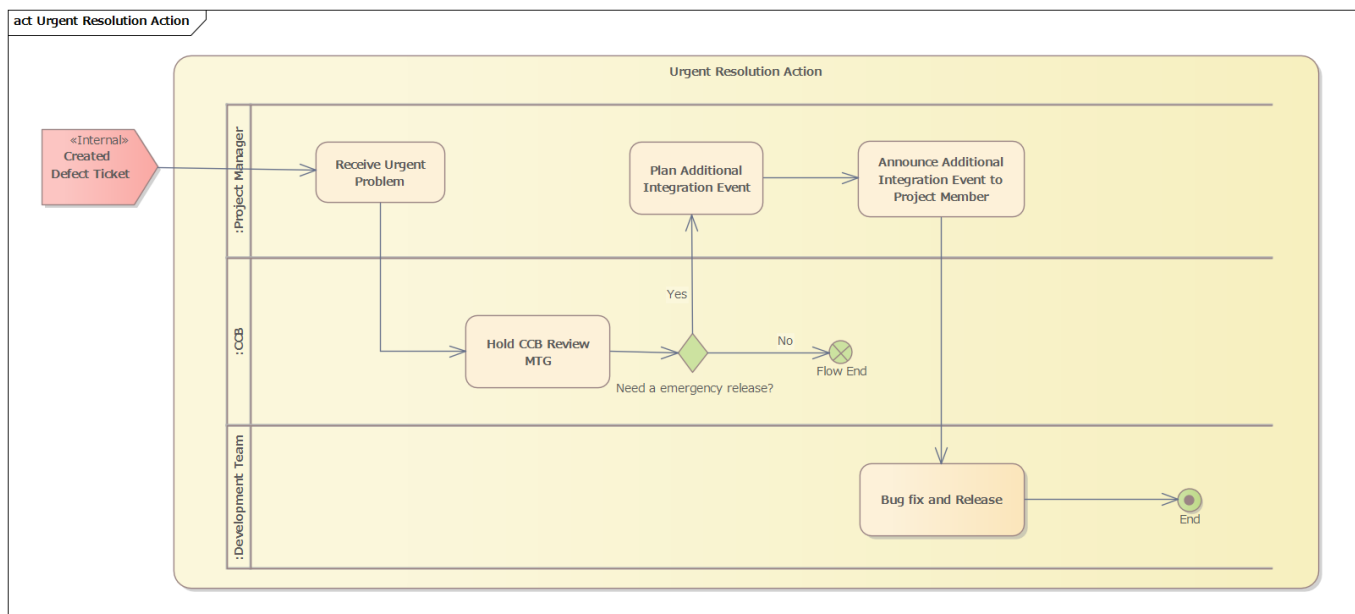
• Trend Analysis

- After Defect Dashboard update, Defect Management analyze trends of defect.
- Defect Management provides the results of trend analysis for the following purpose of use.
 - During the project release cycle
 - To review the project relevant measures and precautions.
 - ex. Whether the project is proceeding as planned. Whether the product meets its quality goals.
 - End the project
 - To improve processes for future projects.

• 傾向分析

- Defect Dashboard を更新し、問題の傾向分析を行う。
- 以下の利用目的とした、傾向分析の結果を提供する。
 - プロジェクトのリリースサイクル中
 - 傾向分析の結果を確認することで、プロジェクトに関連する対策や予防策の検討に繋げる。
 - 例) プロジェクトが計画通りに進行しているかどうか、品質目標が達成できているかどうかなど。
 - プロジェクト終了時
 - 今後のプロジェクトのプロセス改善に繋げる。

6.4.11 Urgent Resolution Action



7 Infrastructure

7.1. Defect Management System

- The usage of a defect Management tool is mandatory for all projects at AlpsAlpine.
- AlpsAlpineでのすべてのプロジェクトでは、不具合管理ツールの使用は必須とする。

7.1.1 Relation with sub-supplier defect management systems

- For managing issues like for example defect a ticket management system is recommended.
 - If the project's ticket management system is not used directly by the supplier,
 - the supplier responsible has to ensure the related tickets are maintained adequately.
 - In that case the supplier responsible has to maintain the tickets in the project's ticket management system by himself, using information collected separately.
 - 不具合などの問題を管理するために、チケット管理システムを利用することを推奨する。
 - もし、プロジェクトのチケット管理システムをサプライヤーが直接利用できない場合、
 - 責任を持つサプライヤーは、関連するチケットが適切に維持されていることを確認しなければならない。
 - 責任を持つサプライヤーは、個別に収集した情報を使用し、プロジェクトのチケット管理システム内のチケットを自身で維持する必要がある。

7.1.2 Relation with Customer defect management systems

- For managing and have a common understanding with customer, use a defect management system to share the contents of the defect ticket with customer is recommended.
- 顧客との共通理解を持ち、不具合管理を行うため、不具合管理システムを使用して、不具合チケットの内容を顧客と共有することを推奨する。

7.2. Trace exchange system

- Storage the traces for the defect and manage the synchronization with customer and subsupplier.
- 不具合のトレースを保管し、顧客とサブサプライヤーとの同期管理を行う。

8 Performance Measurements

- Monitoring (Dashboard Management)
 - Defect Curve
 - TAT
 - Trend analysis
 - Burndown chart
 - Release quality chart

9 Review

- **Planning**
 - Review Defect management plan
- **Executing**
 - Review the Defect management progress
- **Closing**
 - Review the lessons learned.

10 Quality Gates

Defect management quality gate is based on the development phase and the focus point is changing due the Development maturity grade and/or each phases.

* Project preparation phase

- Start analysis the defect process and the infrastructure

* Development phase

- Architecture Phase
 - Create the defect guideline and check the probability of risk for the defect management (including infrastructure).

- Implementation Phase (AKA: Construction Phase)
 - Start running the defect management regularly. (recommend : Once a week)
- Maturity Phase (AKA: Bug-fixing Phase)
 - Running the defect management and ticket review with Customer regularly. (recommend : Once a week)
- Stability-Phase (AKA: Show-stopper Phase, which is too negative)
 - Establish daily review with customer to accelerate to fix the defect and/or discuss to tolerate / shift the ticket.

11 Functional Safety

If functional safety requirement (ASIL) are included in the project scope, execute process which is defined in [FSQSX\[English\]](#) [FSQSX\[Japanese\]](#)

For the defect management scope is mainly Chapter 5.4.3 [FSQSX-MAN\[English\]](#) [FSQSX-MAN\[Japanese\]](#)

For the “3.1 Skills, Capabilities, Responsibility & Authority”, refer [FSQSX-RandR : Roles and Responsibility \[English\]](#) [FSQSX-RandR : Roles and Responsibility \[Japanese\]](#)

12 Cyber Security

If Cyber Security requirement are included in project scope, execute process which is defined in **CSQSX**.

Defect Management Process of CSQSX are defined in the following.
[Regulations](#)

Please see the following Workproducts for Cyber Security.

- [Impact Analysis Result for Change\(CSIAR\)](#)

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